Literature Status: Unique Preduals for Finite Maximal Subdiagonal Algebras

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Status. `literature_already_answered`.

Source question

In Closing Remark 2 on PDF page 21 of [1], Blecher and Labuschagne write:

> A nice question communicated to us by Gilles Godefroy at this conference is whether subdiagonal algebras have unique preduals.

Their standing setting is a finite von Neumann algebra M with faithful normal tracial state τ and a finite maximal subdiagonal algebra $A = H^\infty(M, \tau)$.

Exact later answer

Ueda's introduction explicitly says that his unique-predual result is "an affirmative answer to a question posed by Godefroy stated in" the Blecher–Labuschagne survey [2, p. 2]. His Theorem 3.1, on PDF page 4, states

$$M_*/A_\perp \text{ is the unique predual of } A = H^\infty(M, \tau).$$

This is precisely the canonical predual and exactly the class of algebras in the source question. Thus the answer is yes.

Provenance and scope

The answering author knew he was resolving this exact survey question, so this is an explicit literature answer rather than an agent-derived implication. The packet does not classify preduals outside the finite tracial maximal subdiagonal setting and does not address the other questions in the survey's closing remarks.

References

[1] D. P. Blecher and L. E. Labuschagne, *Von Neumann algebraic H^p theory*, arXiv:math/0611879 (2006/2007).

[2] Y. Ueda, *On peak phenomena for non-commutative H^∞*, arXiv:0802.3449; Math. Ann. **343** (2009), 421–429.